

Week 1: File Permissions & Ownership (Part 1)

Date: March 9, 2026

Study Time: 5-8 hours total

GOAL

Understand rwx permissions and basic chmod/chown usage

STUDY TASKS (1-2 hours)

Reading

☐ "How Linux Works" by Brian Ward

- Chapter 2: Basic Commands and Directory Hierarchy
- Chapter 3: File Modes and Permissions section

Video Learning

☐ LearnLinuxTV - Search: "Linux File Permissions"

- <https://youtube.com/@LearnLinuxTV>

☐ Professor Messer - Search: "Linux+ Permissions"

- <https://youtube.com/@professormesser>

Concepts to Understand

- ☐ Read, Write, Execute (rwx) permissions
 - ☐ chmod numeric notation (755, 644, 600)
 - ☐ chmod symbolic notation (u+rwx, g+rx, o-w)
 - ☐ File ownership (user and group)
 - ☐ Default permissions with umask
-

LAB WORK (3-4 hours)

Setup

```
bash
```

```
# Create practice directory
```

```
mkdir ~/permission-practice
```

```
cd ~/permission-practice
```

Lab Tasks

☐ Create test files

```
bash
```

```
touch file1.txt file2.txt file3.txt
```

```
mkdir dir1 dir2
```

☐ Practice chmod with numeric notation

```
bash
```

```
chmod 755 file1.txt      # rwxr-xr-x
```

```
chmod 644 file2.txt      # rw-r--r--
```

```
chmod 600 file3.txt      # rw-----
```

```
ls -l                    # Verify permissions
```

☐ Practice chmod with symbolic notation

```
bash
```

```
chmod u+x file1.txt      # Add execute for user
```

```
chmod g-w file2.txt      # Remove write for group
```

```
chmod o-r file3.txt      # Remove read for others
```

☐ Create directory structure with different ownership

```
bash
```

```
mkdir shared_folder
```

```
touch shared_folder/team_file.txt
```

```
# Note: You'll need another user account to practice chown
```

☐ Set up proper permissions for a shared folder

```
bash
```

```
chmod 775 shared_folder    # Owner and group can write
ls -ld shared_folder       # Verify directory permissions
```

☐ Test file access with different user accounts

- Try accessing files as different users
- Understand what each permission allows/denies

Verification

```
bash
```

```
# Check your work
ls -l ~/permission-practice
stat file1.txt    # Shows detailed permission info
```

FLASHCARD COMMANDS

Create flashcards for these commands (look up what each does):

1. `chmod`
2. `chown`
3. `chgrp`
4. `ls -l`
5. `stat`
6. `umask`

Flashcard Format:

- Front: Command name
- Back: What it does, syntax, example

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 1.1 Summarize Linux fundamentals
 - 2.4 Given a scenario, manage files and directories
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NOTES & REFLECTIONS

What did you learn this week?

What was challenging?

What commands need more practice?

Questions to research:

Completed: ☐

QUICK REFERENCE

Common Permission Numbers:

- 777 = rwxrwxrwx (everyone can do everything - avoid this!)
- 755 = rwxr-xr-x (standard for executables)
- 644 = rw-r--r-- (standard for files)
- 600 = rw----- (private files)

Permission Symbols:

- r = read (4)
- w = write (2)
- x = execute (1)

User Classes:

- u = user (owner)
 - g = group
 - o = others
 - a = all
-

Next Week: Special Permissions (SUID, SGID, sticky bit)

